(Please write your Enrolment No. immediately) Enrolment No. _____

MID TERM EXAMINATION

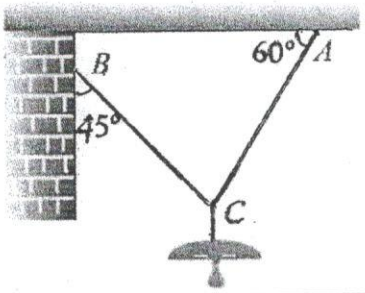
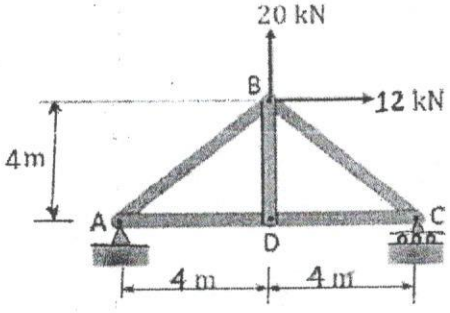
B.TECH PROGRAMMES (UNDER THE AEGIS OF USICT)

2nd Semester, April, 2024

Paper Code: ES 114 **Subject: Engineering Mechanics**

Time: 1½Hrs. **Max. Marks: 30**

Note: Attempt Q.No.1 which is compulsory and any two more questions from remaining.

Q.No.	Question	Max. Marks	CO(s)
1	Write short notes on the following:	10	
(a)	What is the purpose of free body diagram? Explain with a suitable example.	2.5	CO1
(b)	Define centroid, center of mass and center of gravity. Under which conditions is the center of mass different from center of gravity?	2.5	CO1
(c)	What is a truss structure? What is the condition for a perfect truss?	2.5	CO2
(d)	State the laws of dry friction.	2.5	CO2
2	An electric light fixture weighting 15 N hangs from a point C, by two strings AC and BC. The string AC is inclined at 60° to the horizontal and BC at 45° to the horizontal as shown in figure. Determine the forces in the strings AC and BC. 	10	CO1
3	Using the method of joints, determine the axial force in AB, BD and BC members of the truss shown in the figure. 	10	CO2
4	Define angle of fiction and angle of repose. Show that angle of friction is equal to angle of repose for a rough inclined plane.	10	CO2

(b) A straight rod AB, 50cm long has one end B moving with a velocity of 4m/s and the other end A moving along a vertical line YO as shown in figure 8. Find the velocity of the end A and of the midpoint of the rod when it is inclined at 60° with horizontal. **(3)**

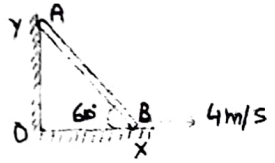


Figure 8

Q9 (a) A reciprocating engine mechanism is shown in figure 9. The crank OA is of length 15cm and rotating at 600 r.p.m. The connecting rod AB is 70cm long. Find the (i) Angular velocity of the connecting rod (ii) The velocity of piston B. **(7)**

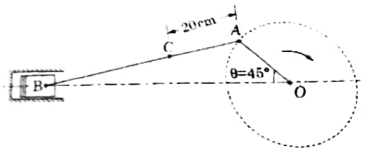


Figure 9

(b) A roller of radius 5.0 cm rides between two horizontal bars moving in the opposite directions as shown in figure 10. Calculate the distance 'd' defining the position of instantaneous centre of rotation of the roller. Assume no slip conditions at the point of contact A and B. **(3)**

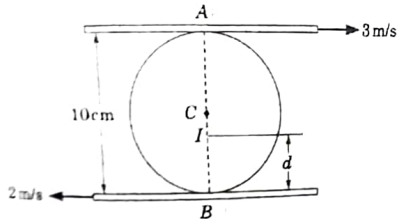


Figure 10

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END TERM EXAMINATION

SECOND SEMESTER [B.TECH] June 2024

Paper Code: ES-114 ***Subject: Engineering Mechanics***

Time: 3 Hours **Maximum Marks: 60**

Note: Attempt all question as directed. Internal choice is indicated.

Q1 Answer the following questions (any **four**) **(4x5=20)**

(a) State the principle of transmissibility of forces.
(b) State the mathematical condition for a rigid or a perfect truss.
(c) State D'Alembert's principle.
(d) What is the point of contraflexure in case of a beam.
(e) Define instantaneous centre of rotation.
(f) What is a self locking condition in case of a machine.

UNIT-I

Q2 (a) State and proof Varignon's Theorem. **(3)**

(b) Find the moment of inertia of the cross section of the iron beam (Fig.1) with respect to the centroidal axis. **(7)**

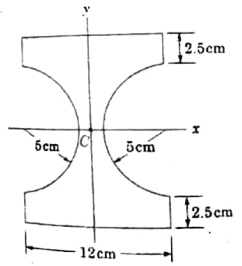


Figure 1

Q3 (a) State Pappus Theorem with an example. **(1)**

(b) State the prove parallel axis theorem. **(3)**

(c) Two equal weights each of 1000N is supported by a flexible string (Fig. 2). Find the tensions in the portion AB, BC and CD of the string **(6)**

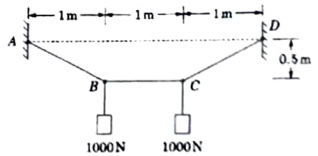


Figure 2

P.T.O.